BUSINESS INTELLIGENCE · PORTFOLIO CASE STUDY

Sales & Marketing Call Center Dashboard

I built an interactive Excel dashboard to understand where calls were being lost, how agent workload related to outcomes, and which operational issues deserved attention first.

200,000	41.8%	58.2%
CALL RECORDS	**SUCCESS RATE**	**UNSUCCESSFUL**
12-month operational dataset	83,519 successful calls	failure + abandoned

THE BUSINESS QUESTION

Where are unsuccessful calls concentrated, what is driving them, and how can a manager move from a high-level KPI to an agent-level explanation?

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Microsoft Excel · Power Query · PivotTables · PivotCharts · Slicers

01 · PROJECT FRAMING

From raw call records to a management view

I did not want the dashboard to be a collection of charts. I designed it around a simple decision flow: start with the overall outcome, identify where the gap is concentrated, then drill into an individual agent by time, location, product and reason.

DATA I WORKED WITH

- 200,000 call records
- 18 source fields
- 13 call center agents
- 5 product categories
- Customer location, demographics and time of day
- Outcome, call duration, rating, follow-up and repeat-customer fields

HOW I BUILT IT

- Loaded the Dataset query through Power Query
- Used PivotTables to build reusable aggregations
- Connected PivotCharts to the analysis layer
- Added slicers and an agent selector for drill-down
- Kept the dashboard focused on business questions rather than raw fields

The questions behind the dashboard

- What share of calls end in success, failure or abandonment?
- Which agents handle the most volume, and how do their outcomes compare?
- What reasons appear most often when a call is unsuccessful?
- Which products are most common among successful calls?
- For one selected agent, how do volume, geography and unsuccessful-call reasons change the story?

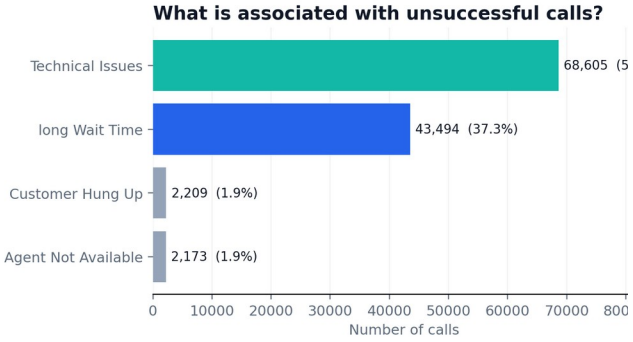
> **WHY THIS STRUCTURE MATTERS**
> A manager can move from a 200,000-call headline to a specific operational area without opening the raw data or rebuilding a report.

02 · EXECUTIVE RESULTS

The main issue is not call volume. It is what happens to the calls.

Across the full dataset, 83,519 calls were successful. The remaining 116,481 calls ended in failure or abandonment, so the overall success rate was 41.8%.





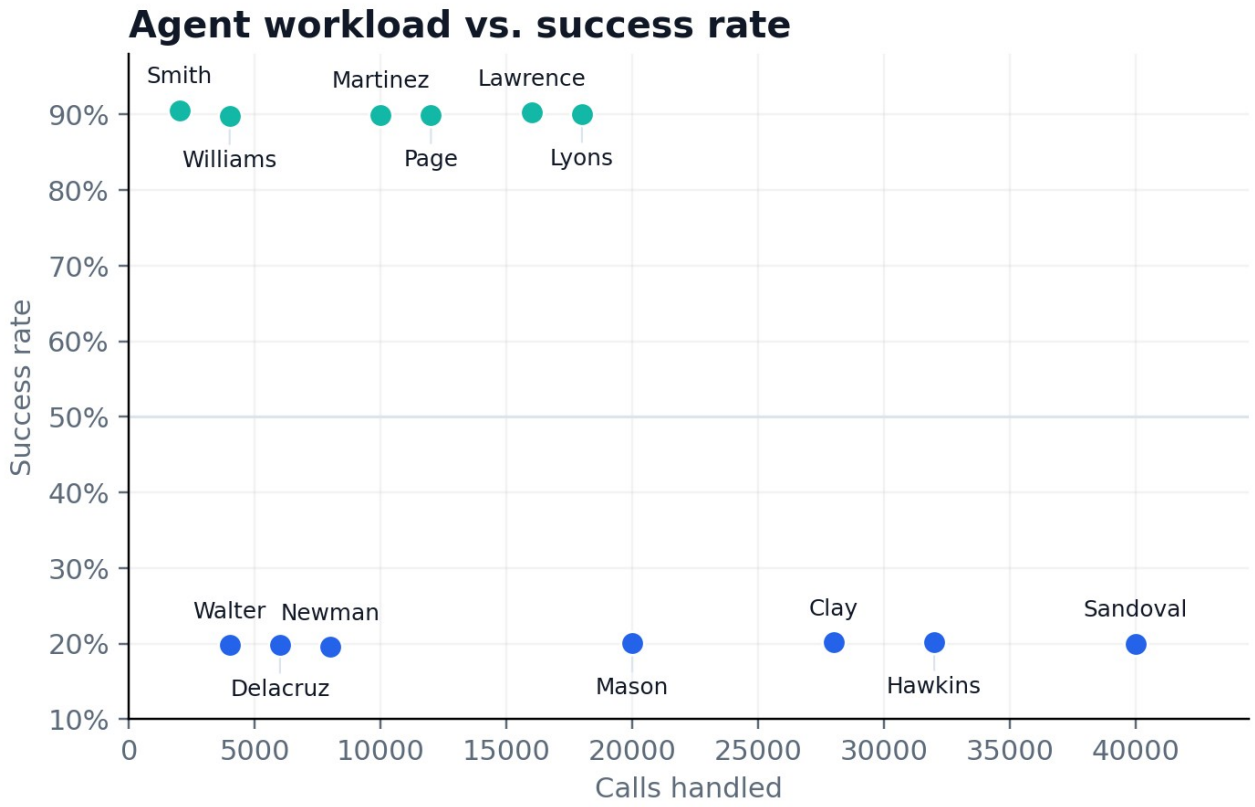
THE MOST ACTIONABLE FINDING

Technical Issues account for 58.9% of unsuccessful calls and Long Wait Time another 37.3%. Together they explain 96.2% of the unsuccessful-call reason records.

That split changes the type of action I would recommend. The first problem points to technical or process reliability; the second points to capacity, routing or queue management. “Customer Hung Up” and “Agent Not Available” are much smaller by comparison.

The gap is highly concentrated in the busiest part of the team

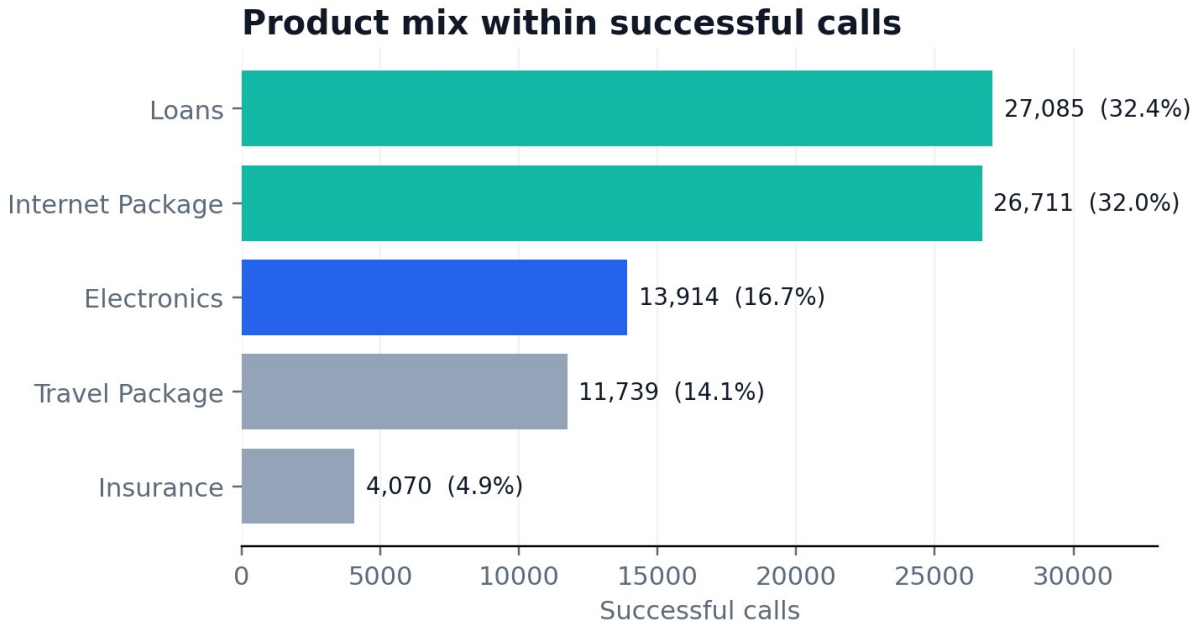
The agent view revealed a strong concentration pattern. The four highest-volume agents - Ava Sandoval, Elijah Hawkins, Drew Clay and Ray Mason - handled 120,000 calls, equal to 60.0% of total volume.



60%	82.3%	20.1%
OF ALL CALLS	**OF UNSUCCESSFUL CALLS**	**SUCCESS RATE**
handled by the four busiest agents	come from those same four agents	across those four agents

This is a priority signal, not proof that the agents themselves are the cause. Before treating it as a coaching issue, I would compare their product mix, call complexity, time-of-day load and routing rules. If the case mix is comparable, then coaching and process review become much stronger next steps.

Successful calls are concentrated in two product categories



Loans (27,085) and Internet Package (26,711) together represent 64.4% of successful calls. This gives the dashboard a useful commercial lens: once the operational outcome is understood, managers can see where successful conversations are concentrated.

A DATA-DEFINITION CHECK I WOULD KEEP IN THE REPORT

The workbook field is named “Reason_Call_Abandoned”, but its counts add to 116,481 - exactly Failure + Abandoned, not Abandoned alone. I therefore describe it as an unsuccessful-call reason in this report. This avoids turning a field name into a misleading business claim.

A second boundary I kept explicit

The product pivot is filtered to successful calls: the five product counts sum exactly to 83,519 successful calls. I use it as a successful-product mix, not as the product distribution of all 200,000 calls.

WHY I INCLUDED THESE NOTES

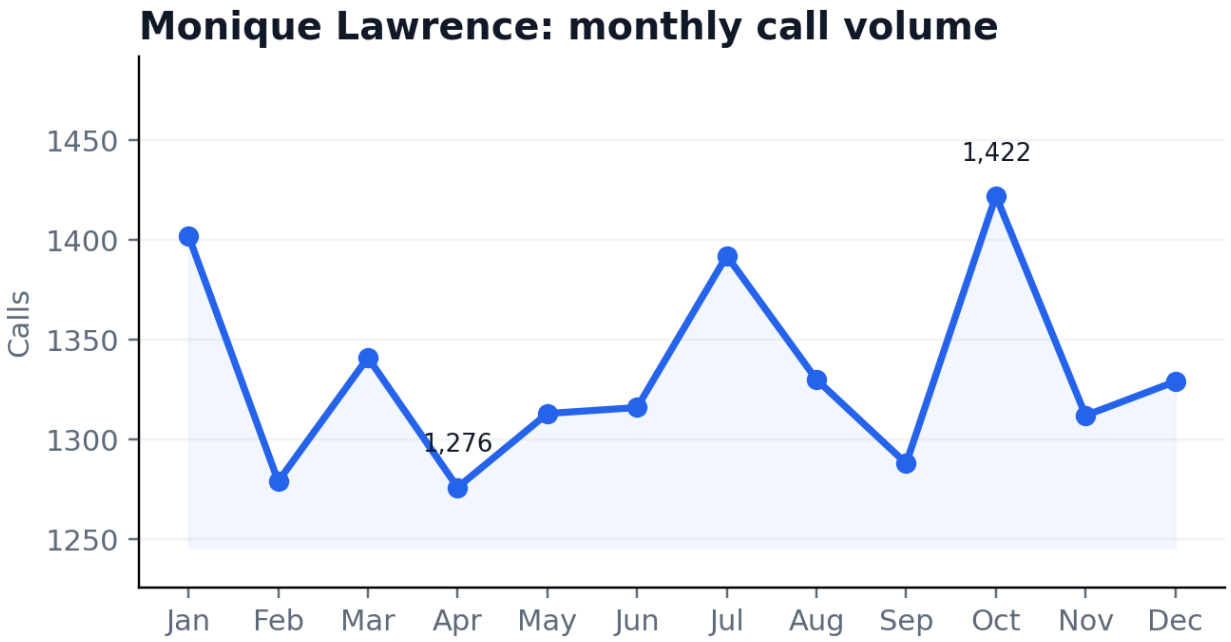
Good dashboards are not only about attractive visuals. The definitions behind each KPI have to reconcile with the data, especially when the report may be viewed without the workbook open.

05 · INTERACTIVE DRILL-DOWN

Example: Monique Lawrence

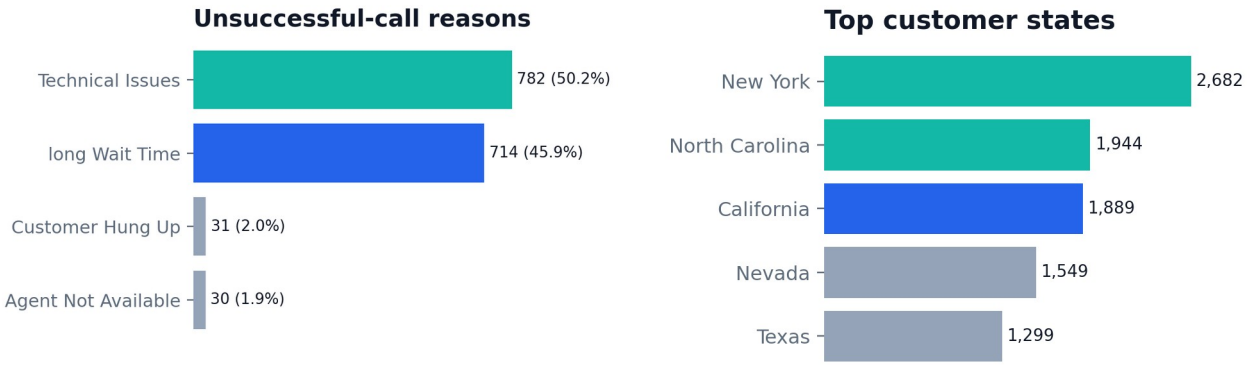
The dashboard lets me switch from the full call center view to one agent. The current workbook selection is Monique Lawrence, so I use that view here as an example of the drill-down logic.

16,000	90.3%	4.25/5	16.43 min
TOTAL CALLS	SUCCESS RATE	AVG. RATING	AVG. DURATION
8.0% of all calls	14,443 successful calls	customer rating	per call



Her monthly workload is relatively stable: the low point is April at 1,276 calls and the high point is October at 1,422, around an average of 1,333 calls per month. That stability makes the outcome and reason mix easier to interpret without a large seasonal spike dominating the view.

For this agent, the same two operational themes still dominate



Among Monique Lawrence's 1,557 unsuccessful calls, Technical Issues account for 782 (50.2%) and Long Wait Time for 714 (45.9%). Her top five states account for 58.5% of her calls, led by New York (2,682).

What I would recommend next

- Start with a root-cause review of Technical Issues, because this is the largest unsuccessful-call reason at both the overall and selected-agent level.
- Review queue capacity and staffing by time of day to address Long Wait Time, the second-largest driver.
- Investigate the four high-volume / low-success agents by case mix before deciding whether the issue is coaching, routing, workload or a combination of them.
- Protect the high-performing pattern by documenting how the ~90% success-rate agents handle comparable calls and turning repeatable practices into coaching material.
- Use the agent selector and slicers as a recurring management review rather than treating the dashboard as a one-time presentation.

07 · CONCLUSION

What this project demonstrates

This project shows how I approach analytics work: I start with the business question, build a clear reporting layer, check whether KPI definitions reconcile with the underlying totals, and only then turn the analysis into recommendations.

Power Query	PivotTables	Dashboard
PREPARE	ANALYZE	COMMUNICATE
reusable dataset connection	consistent aggregations	interactive management view

What I would improve in a production version

- Add a clear operational definition for “success”, “failure” and “abandoned” so each outcome has a documented business rule.
- Add service-level metrics such as queue time, first-call resolution and SLA attainment if those fields become available.
- Separate call-complexity or campaign effects from agent performance before using the dashboard for individual evaluation.
- Automate refresh and publish a controlled version for recurring management use.
- Add statistical or cohort checks where comparisons could be affected by unequal call mix.

> **PORTFOLIO TAKEAWAY**
>
> The dashboard is not valuable because it contains 200,000 rows. It is valuable because it turns those rows into a short chain of decisions: see the gap, locate the concentration, understand the reason, then decide where to investigate.

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